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Materials ML Failure Analysis

Public Research Compendium v1.0.0 | Zenodo DOI: 10.5281/zenodo.21919563

Three empirical disagreements about what materials-machine-learning evidence can safely mean

Failure analysis -> interpretation -> viability -> authority

0.0394	73.3% -> 4.51%	927,361 max delta = 0
Fitted use vs retrained dependence	Non-null signal vs absolute viability	Exact replay vs bounded authority

What this compendium is

A public evidence package containing three papers, a programme monograph, a public technical companion, accepted derived tables, replay summaries and small scripts that reconstruct the principal reported readouts.

What it is not

It is not the production Systemica/Geatomica runtime. It does not publish reusable orchestration, authority-control, memory, progression, recovery, routing or deployment implementations. It does not grant engineering permission.

OPEN CLAIMS	OPEN EVIDENCE	MINIMAL REPRODUCTION	PRIVATE REUSABLE MACHINERY

Evidence ceiling

The package reports bounded local measurement and replay. It does not establish external replication, peer review, causal mechanism, physical qualification, universal thresholds or automatic engineering deployment.

01 | The scientific trilogy

Three papers, three non-interchangeable evidence objects

Paper 1	**Importance Is Not Necessity**	Under one frozen Matbench/Magpie/MLP protocol, fitted permutation importance and complete retrained-removal consequence produced weakly agreeing descriptor accounts. The full 3,440-unit route repeated exactly.
Paper 2	**Better Than Null Is Not Yet Useful**	In 288 paired-null-comparable Materials Project elastic cells, 73.3% beat the paired null while only 4.51% cleared the minimal absolute R2 boundary. Relative signal and task viability were therefore kept as separate gates.
Paper 3	**Available Evidence Is Not Engineering Authority**	Admission, viability, replay and bounded action can fail independently. Exact replay proves that a decision repeated; it does not automatically strengthen a held or conditional scientific claim.

The shared rule

No one evidence object should silently impersonate a stronger one: fitted importance is not retrained dependence; non-null signal is not useful prediction; completed computation is not replay; replayed evidence is not engineering permission.

Materials context

JARVIS, Materials Project, Matbench and spatial representations provide different witnesses. Their populations, models, controls and replay states are kept separate. Adverse and conditional outcomes remain visible rather than being averaged into a single programme PASS.

Read next

Begin with the paper matching your question. Use the monograph for the full scientific lineage and the Technical & Industrial Companion for the public runtime, recovery, memory and deployment boundary.

02 | Evidence and replay

What repeated, and what did not become stronger

Route	Measured/replayed object	Bounded reading
JARVIS	48 fresh + 48 replay; 28,248 numeric cells; max delta about 1.07e-14	Numerically equivalent at frozen tolerance; raw-byte divergence remains disclosed.
Materials Project	64 fresh + 64 replay; 37,924 cells; 916 strict exceedances, all elastic	Admission and elastic viability remain held or conditional; adverse replay evidence is retained.
Matbench V0.3	3,440 fresh + 3,440 repeat; 65,740 cells; zero differences	Exact same-environment repeat of the declared descriptor-response route.
Spatial V0.6.1	576 fresh + 576 replay; 927,361 cells; byte, metadata and numerical identity	Bulk metric geometry passed both views; shear polarity remained conditional because one required view failed.

Fresh, resume and replay are different

Fresh creates a new execution. Resume continues the same verified identity from accepted checkpoints. Replay separately re-executes a frozen plan under a new identity and compares declared contracts. Recovery protects continuity; replay measures repeatability.

Scientific memory boundary

Source memory, execution memory, scientific evidence memory and narrative working memory serve different roles. Narrative summaries can help reconstruct the work, but cannot overwrite accepted scientific evidence or promote a held result.

03 | How to inspect the release

A small public surface with explicit limits

1	Read	Use Papers 1-3 for the falsifiable findings, the monograph for programme lineage and the public companion for implementation and industrial context.
2	Reconstruct	Run the three standard-library scripts in reproduction/. They read accepted public CSV/JSON tables and reconstruct headline statistics without fitting models.
3	Verify	Check MANIFEST.sha256, replay summaries, claim boundaries and QA. The archive contains no Git history, private runtime source or raw third-party corpus.
4	Challenge	Treat the limitations and falsification routes as part of the result. External replication, alternative models, tensor-resolved elasticity and physical validation remain open work.

Public implementation boundary

The release opens claims, evidence and minimal claim reconstruction. Reusable progression policy, orchestration, recovery implementation, authority engines, memory backends, worker routing and production integration remain outside the package.

Third-party data

Raw JARVIS, Materials Project and Matbench corpora are not redistributed. The release contains derived accepted tables and retrieval/citation context; provider terms remain controlling.

Release and citation

This is the public v1.0.0 research compendium archived under Zenodo DOI 10.5281/zenodo.21919563. The release supports scholarly inspection; it is not peer review, external validation, physical qualification or engineering permission.

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